

APPLICATION OF DERIVATIVES

Exercise-1 : Single Choice Problems

- The difference between the maximum and minimum value of the function $f(x) = 3\sin^4 x - \cos^6 x$ is :
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 4
- A function $y = f(x)$ has a second order derivative $f''(x) = 6(x-1)$. If its graph passes through the point (2, 1) and at that point the tangent to the graph is $y = 3x - 5$, then the function is :
 (a) $(x-1)^2$ (b) $(x-1)^3$ (c) $(x+1)^3$ (d) $(x+1)^2$
- If the subnormal at any point on the curve $y = 3^{1-k} \cdot x^k$ is of constant length then k equals to :
 (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 0
- If $x^5 - 5qx + 4r$ is divisible by $(x-c)^2$ then which of the following must hold true $\forall q, r, c \in \mathbb{R}$?
 (a) $q = r$ (b) $q + r = 0$ (c) $q^5 = r^4$ (d) $q^4 = r^5$
- A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is :
 (a) $\frac{1}{36\pi} \text{ cm/min}$ (b) $\frac{1}{18\pi} \text{ cm/min}$ (c) $\frac{1}{54\pi} \text{ cm/min}$ (d) $\frac{5}{6\pi} \text{ cm/min}$
- If $f(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$, then number of local extremas for $g(x)$, where $g(x) = f(|x|)$:
 (a) 3 (b) 4 (c) 5 (d) None of these
- Two straight roads OA and OB intersect at an angle 60° . A car approaches O from A, where $OA = 700 \text{ m}$ at a uniform speed of 20 m/s, Simultaneously, a runner starts running from O towards B at a uniform speed of 5 m/s. The time after start when the car and the runner are closest is :
 (a) 10 sec (b) 15 sec
 (c) 20 sec (d) 30 sec

8. Let $f(x) = \begin{cases} a-3x & ; -2 \leq x < 0 \\ 4x+3 & ; 0 \leq x < 1 \end{cases}$; if $f(x)$ has smallest value at $x = 0$, then range of a , is :
- (a) $(-\infty, 3)$ (b) $(-\infty, 3]$ (c) $(3, \infty)$ (d) $[3, \infty)$
9. $f(x) = \begin{cases} 3+|x-k| & , x \leq k \\ a^2 - 2 + \frac{\sin(x-k)}{(x-k)} & , x > k \end{cases}$ has minimum at $x = k$, then :
- (a) $a \in \mathbb{R}$ (b) $|a| < 2$ (c) $|a| > 2$ (d) $1 < |a| < 2$
10. For a certain curve $\frac{d^2y}{dx^2} = 6x - 4$ and curve has local minimum value 5 at $x = 1$. Let the global maximum and global minimum values, where $0 \leq x \leq 2$; are M and m . Then the value of $(M - m)$ equals to :
- (a) -2 (b) 2 (c) 12 (d) -12
11. The tangent to $y = ax^2 + bx + \frac{7}{2}$ at $(1, 2)$ is parallel to the normal at the point $(-2, 2)$ on the curve $y = x^2 + 6x + 10$. Then the value of $\frac{a}{2} - b$ is :
- (a) 2 (b) 0 (c) 3 (d) 1
12. If (a, b) be the point on the curve $9y^2 = x^3$ where normal to the curve make equal intercepts with the axis, then the value of $(a + b)$ is :
- (a) 0 (b) $\frac{10}{3}$ (c) $\frac{20}{3}$ (d) None of these
13. The curve $y = f(x)$ satisfies $\frac{d^2y}{dx^2} = 6x - 4$ and $f(x)$ has a local minimum value 5 when $x = 1$. Then $f(0)$ is equal to :
- (a) 1 (b) 0 (c) 5 (d) None of these
14. Let A be the point where the curve $5\alpha^2x^3 + 10\alpha x^2 + x + 2y - 4 = 0$ ($\alpha \in \mathbb{R}, \alpha \neq 0$) meets the y -axis, then the equation of tangent to the curve at the point where normal at A meets the curve again, is :
- (a) $x - \alpha y + 2\alpha = 0$ (b) $\alpha x + y - 2 = 0$ (c) $2x - y + 2 = 0$ (d) $x + 2y - 4 = 0$
15. The difference between the greatest and the least value of the function $f(x) = \cos x + \frac{1}{2} \cos 2x - \frac{1}{3} \cos 3x$
- (a) $\frac{11}{5}$ (b) $\frac{13}{6}$ (c) $\frac{9}{4}$ (d) $\frac{7}{3}$
16. The x -co-ordinate of the point on the curve $y = \sqrt{x}$ which is closest to the point $(2, 1)$ is :
- (a) $\frac{2+\sqrt{3}}{2}$ (b) $\frac{1+\sqrt{3}}{2}$ (c) $\frac{-1+\sqrt{3}}{2}$ (d) 1

17. The tangent at a point P on the curve $y = \ln \left(\frac{2 + \sqrt{4 - x^2}}{2 - \sqrt{4 - x^2}} \right) - \sqrt{4 - x^2}$ meets the y -axis at T ; then

PT^2 equals to :

- (a) 2 (b) 4 (c) 8 (d) 16

18. Let $f(x) = \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 1$

and $g(x) = \int_1^x (2t^2 - \ln t) f(t) dt$ ($x > 1$), then :

- (a) g is increasing on $(1, \infty)$
 (b) g is decreasing on $(1, \infty)$
 (c) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
 (d) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

19. Let $f(x) = x^3 + 6x^2 + ax + 2$, if $(-3, -1)$ is the largest possible interval for which $f(x)$ is decreasing function, then $a =$

- (a) 3 (b) 9 (c) -2 (d) 1

20. Let $f(x) = \tan^{-1} \left(\frac{1-x}{1+x} \right)$. Then difference of the greatest and least value of $f(x)$ on $[0, 1]$ is :

- (a) $\pi/2$ (b) $\pi/4$ (c) π (d) $\pi/3$

21. The number of integral values of a for which $f(x) = x^3 + (a+2)x^2 + 3ax + 5$ is monotonic in $\forall x \in \mathbb{R}$.

- (a) 2 (b) 4 (c) 6 (d) 7

22. The number of critical points of $f(x) = \left(\int_0^x (\cos^2 t - \sqrt[3]{t}) dt \right) + \frac{3}{4}x^{4/3} - \frac{x+1}{2}$ in $[0, 6\pi]$ is :

- (a) 10 (b) 8 (c) 6 (d) 12

23. Let $f(x) = \min \left(\frac{1}{2} - \frac{3x^2}{4}, \frac{5x^2}{4} \right)$ for $0 \leq x \leq 1$, then maximum value of $f(x)$ is :

- (a) 0 (b) $\frac{5}{64}$
 (c) $\frac{5}{4}$ (d) $\frac{5}{16}$

24. Let $f(x) = \begin{cases} 2 - |x^2 + 5x + 6| & x \neq -2 \\ b^2 + 1 & x = -2 \end{cases}$

Has relative maximum at $x = -2$, then complete set of values b can take is :

- (a) $|b| \geq 1$ (b) $|b| < 1$ (c) $b > 1$ (d) $b < 1$

25. Let for the function $f(x) = \begin{cases} \cos^{-1} x & ; -1 \leq x \leq 0, \\ mx + c & ; 0 < x \leq 1 \end{cases}$

Lagrange's mean value theorem is applicable in $[-1, 1]$ then ordered pair (m, c) is :

- (a) $\left(1, -\frac{\pi}{2}\right)$ (b) $\left(1, \frac{\pi}{2}\right)$ (c) $\left(-1, -\frac{\pi}{2}\right)$ (d) $\left(-1, \frac{\pi}{2}\right)$
26. Tangents are drawn to $y = \cos x$ from origin then points of contact of these tangents will always lie on :
- (a) $\frac{1}{x^2} = \frac{1}{y^2} + 1$ (b) $\frac{1}{x^2} = \frac{1}{y^2} - 2$ (c) $\frac{1}{y^2} = \frac{1}{x^2} + 1$ (d) $\frac{1}{y^2} = \frac{1}{x^2} - 2$
27. Least natural number a for which $x + ax^{-2} > 2 \forall x \in (0, \infty)$ is :
- (a) 1 (b) 2 (c) 5 (d) None of these
28. Angle between the tangents to the curve $y = x^2 - 5x + 6$ at points $(2, 0)$ and $(3, 0)$ is :
- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
29. Difference between the greatest and least values of the function $f(x) = \int_0^x (\cos^2 t + \cos t + 2) dt$ in the interval $[0, 2\pi]$ is $K\pi$, then K is equal to :
- (a) 1 (b) 3 (c) 5 (d) None of these
30. The range of the function $f(\theta) = \frac{\sin \theta}{\theta} + \frac{\theta}{\tan \theta}$, $\theta \in \left(0, \frac{\pi}{2}\right)$ is equal to :
- (a) $(0, \infty)$ (b) $\left(\frac{1}{\pi}, 2\right)$ (c) $(2, \infty)$ (d) $\left(\frac{2}{\pi}, 2\right)$
31. Number of integers in the range of c so that the equation $x^3 - 3x + c = 0$ has all its roots real and distinct is :
- (a) 2 (b) 3 (c) 4 (d) 5
32. Let $f(x) = \int e^x (x-1)(x-2) dx$. Then $f(x)$ decreases in the interval :
- (a) $(2, \infty)$ (b) $(-2, -1)$
(c) $(1, 2)$ (d) $(-\infty, 1) \cup (2, \infty)$
33. If the cubic polynomial $y = ax^3 + bx^2 + cx + d$ ($a, b, c, d \in R$) has only one critical point in its entire domain and $ac = 2$, then the value of $|b|$ is :
- (a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $\sqrt{5}$ (d) $\sqrt{6}$
34. On the curve $y = \frac{1}{1+x^2}$, the point at which $\left|\frac{dy}{dx}\right|$ is greatest in the first quadrant is :
- (a) $\left(\frac{1}{2}, \frac{4}{5}\right)$ (b) $\left(1, \frac{1}{2}\right)$ (c) $\left(\frac{1}{\sqrt{2}}, \frac{2}{3}\right)$ (d) $\left(\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$

35. If $f(x) = 2x$, $g(x) = 3 \sin x - x \cos x$, then for $x \in \left(0, \frac{\pi}{2}\right)$:
- (a) $f(x) > g(x)$ (b) $f(x) < g(x)$
 (c) $f(x) = g(x)$ has exactly one real root. (d) $f(x) = g(x)$ has exactly two real roots
36. Let $f(x) = \sin^{-1} \left(\frac{2g(x)}{1+g(x)^2} \right)$, then which are correct?
- (i) $f(x)$ is decreasing if $g(x)$ is increasing and $|g(x)| > 1$
 (ii) $f(x)$ is an increasing function if $g(x)$ is increasing and $|g(x)| \leq 1$
 (iii) $f(x)$ is decreasing function if $g(x)$ is decreasing and $|g(x)| > 1$
 (a) (i) and (iii) (b) (i) and (ii) (c) (i), (ii) and (iii) (d) (iii)
37. The graph of the function $y = f(x)$ has a unique tangent at $(e^a, 0)$ through which the graph passes then $\lim_{x \rightarrow e^a} \frac{\ln(1+7f(x)) - \sin(f(x))}{3f(x)}$ is equal to :
- (a) 1 (b) 3 (c) 2 (d) 7
38. Let $f(x)$ be a function such that $f'(x) = \log_{1/3}(\log_3(\sin x + a))$. The complete set of values of 'a' for which $f(x)$ is strictly decreasing for all real values of x is :
- (a) $[4, \infty)$ (b) $[3, 4]$ (c) $(-\infty, 4)$ (d) $[2, \infty)$
39. If $f(x) = a \ln|x| + bx^2 + x$ has extremas at $x = 1$ and $x = 3$, then :
- (a) $a = \frac{3}{4}, b = -\frac{1}{8}$ (b) $a = \frac{3}{4}, b = \frac{1}{8}$ (c) $a = -\frac{3}{4}, b = -\frac{1}{8}$ (d) $a = -\frac{3}{4}, b = \frac{1}{8}$
40. Let $f(x) = \begin{cases} 1 + \sin x, & x < 0 \\ x^2 - x + 1, & x \geq 0 \end{cases}$, then :
- (a) f has a local maximum at $x = 0$ (b) f has a local minimum at $x = 0$
 (c) f is increasing everywhere (d) f is decreasing everywhere
41. If m and n are positive integers and $f(x) = \int_1^x (t-a)^{2n}(t-b)^{2m+1} dt$, $a \neq b$, then :
- (a) $x = b$ is a point of local minimum (b) $x = b$ is a point of local maximum
 (c) $x = a$ is a point of local minimum (d) $x = a$ is a point of local maximum
42. For any real θ , the maximum value of $\cos^2(\cos \theta) + \sin^2(\sin \theta)$ is :
- (a) 1 (b) $1 + \sin^2 1$
 (c) $1 + \cos^2 1$ (d) Does not exist
43. If the tangent at P of the curve $y^2 = x^3$ intersects the curve again at Q and the straight line OP, OQ have inclinations α, β where O is origin, then $\left(\frac{\tan \alpha}{\tan \beta} \right)$ has the value, equals to :
- (a) -1 (b) -2 (c) 2 (d) $\sqrt{2}$

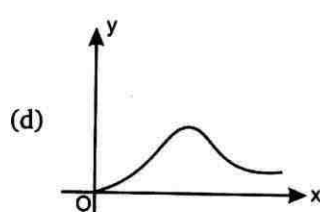
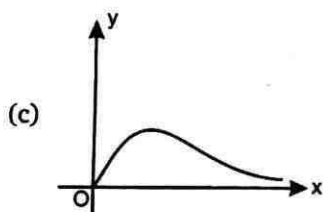
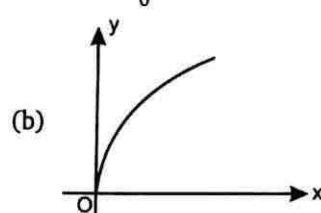
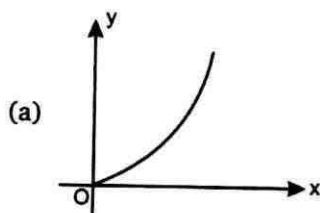
44. If $x + 4y = 14$ is a normal to the curve $y^2 = \alpha x^3 - \beta$ at $(2, 3)$, then value of $\alpha + \beta$ is :

- (a) 9 (b) -5 (c) 7 (d) -7

45. The tangent to the curve $y = e^{kx}$ at a point $(0, 1)$ meets the x -axis at $(a, 0)$ where $a \in [-2, -1]$, then $k \in$:

- (a) $[-\frac{1}{2}, 0]$ (b) $[-1, -\frac{1}{2}]$ (c) $[0, 1]$ (d) $[\frac{1}{2}, 1]$

46. Which of the following graph represent the function $f(x) = \int_0^{\sqrt{x}} e^{-\frac{u^2}{x}} du$, for $x > 0$ and $f(0) = 0$?



47. Let $f(x) = (x-a)(x-b)(x-c)$ be a real valued function where $a < b < c$ ($a, b, c \in \mathbb{R}$) such that $f''(\alpha) = 0$. Then if $\alpha \in (c_1, c_2)$, which one of the following is correct ?

- (a) $a < c_1 < b$ and $b < c_2 < c$ (b) $a < c_1, c_2 < b$
(c) $b < c_1, c_2 < c$ (d) None of these

48. $f(x) = x^6 - x - 1$, $x \in [1, 2]$. Consider the following statements :

- (1) f is increasing on $[1, 2]$ (2) f has a root in $[1, 2]$
(3) f is decreasing on $[1, 2]$ (4) f has no root in $[1, 2]$

Which of the above are correct?

- (a) 1 and 2 (b) 1 and 4 (c) 2 and 3 (d) 3 and 4

49. Which one of the following curves is the orthogonal trajectory of straight lines passing through a fixed point (a, b) ?

- (a) $x - a = k(y - b)$ (b) $(x - a)(y - b) = k$
(c) $(x - a)^2 = k(y - b)$ (d) $(x - a)^2 + (y - b)^2 = k$

50. The function $f(x) = \sin^3 x - m \sin x$ is defined on open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and if assumes only 1 maximum value and only 1 minimum value on this interval. Then, which one of the following must be correct ?
 (a) $0 < m < 3$ (b) $-3 < m < 0$ (c) $m > 3$ (d) $m < -3$
51. The greatest of the numbers $1, 2^{1/2}, 3^{1/3}, 4^{1/4}, 5^{1/5}, 6^{1/6}$ and $7^{1/7}$ is :
 (a) $2^{1/2}$ (b) $3^{1/3}$ (c) $7^{1/7}$ (d) $6^{1/6}$
52. Let l be the line through $(0, 0)$ and tangent to the curve $y = x^3 + x + 16$. Then the slope of l equal to :
 (a) 10 (b) 11 (c) 17 (d) 13
53. The slope of the tangent at the point of inflection of $y = x^3 - 3x^2 + 6x + 2009$ is equal to :
 (a) 2 (b) 3 (c) 1 (d) 4
54. Let f be a real valued function with $(n+1)$ derivatives at each point of R . For each pair of real numbers $a, b, a < b$, such that

$$\ln \left[\frac{f(b) + f'(b) + \dots + f^{(n)}(b)}{f(a) + f'(a) + \dots + f^{(n)}(a)} \right] = b - a$$

Statement-1 : There is a number $c \in (a, b)$ for which $f^{(n+1)}(c) = f(c)$

because

Statement-2 : If $h(x)$ be a derivable function such that $h(p) = h(q)$ then by Rolle's theorem $h'(d) = 0, d \in (p, q)$

- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
 (b) Statement-1 is true, statement-2 is true and statement-2 is not correct explanation for statement-1
 (c) Statement-1 is true, statement-2 is false
 (d) Statement-1 is false, statement-2 is true
55. If $g(x)$ is twice differentiable real valued function satisfying $g''(x) - 3g'(x) > 3 \forall x \geq 0$ and $g'(0) = -1$, then $h(x) = g(x) + x \forall x > 0$ is :
 (a) strictly increasing (b) strictly decreasing
 (c) non monotonic (d) data insufficient
56. If the straight line joining the points $(0, 3)$ and $(5, -2)$ is tangent to the curve $y = \frac{c}{x+1}$; then the value of c is :
 (a) 2 (b) 3 (c) 4 (d) 5
57. Number of solutions(s) of $\ln |\sin x| = -x^2$ if $x \in \left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is/are :
 (a) 2 (b) 4 (c) 6 (d) 8

58. The equation $\sin^{-1} x = |x - a|$ will have atleast one solution then complete set of values of a be :
- (a) $[-1, 1]$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (c) $\left[1 - \frac{\pi}{2}, 1 + \frac{\pi}{2}\right]$ (d) $\left[\frac{\pi}{2} - 1, \frac{\pi}{2} + 1\right]$
59. For any real number b , let $f(b)$ denotes the maximum of $\left|\sin x + \frac{2}{3 + \sin x} + b\right| \forall x \in R$.
- Then the minimum value of $f(b) \forall b \in R$ is :
- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 1
60. Which of the following are correct
- (a) $x^4 + 2x^2 - 6x + 2 = 0$ has exactly four real solution
- (b) $x^5 + 5x + 1 = 0$ has exactly three real solutions
- (c) $x^n + ax + b = 0$ where n is an even natural number has atmost two real solution $a, b, \in R$.
- (d) $x^3 - 3x + c = 0, c > 0$ has two real solution for $x \in (0, 1)$
61. For any real number b , let $f(b)$ denotes the maximum of $\left|\sin x + \frac{2}{3 + \sin x} + b\right| \forall x \in R$. Then the minimum value of $f(b) \forall b \in R$ is :
- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 1
62. If p be a point on the graph of $y = \frac{x}{1 + x^2}$, then coordinates of 'p' such that tangent drawn to curve at p has the greatest slope in magnitude is :
- (a) $(0, 0)$ (b) $\left(\sqrt{3}, \frac{\sqrt{3}}{4}\right)$ (c) $\left(-\sqrt{3}, -\frac{\sqrt{3}}{4}\right)$ (d) $\left(1, \frac{1}{2}\right)$
63. Let $f: [0, 2\pi] \rightarrow [-3, 3]$ be a given function defined as $f(x) = 3 \cos \frac{x}{2}$. The slope of the tangent to the curve $y = f^{-1}(x)$ at the point where the curve crosses the y -axis is :
- (a) -1 (b) $-\frac{2}{3}$ (c) $-\frac{1}{6}$ (d) $-\frac{1}{3}$
64. Number of stationary points in $[0, \pi]$ for the function $f(x) = \sin x + \tan x - 2x$ is :
- (a) 0 (b) 1 (c) 2 (d) 3
65. If $a, b, c, d \in R$ such that $\frac{a + 2c}{b + 3d} + \frac{4}{3} = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has
- (a) atleast one root in $(-1, 0)$ (b) atleast one root in $(0, 1)$
- (c) no root in $(-1, 1)$ (d) no root in $(0, 2)$

66. If $f'(x) = \phi(x)(x-2)^2$. Where $\phi(2) \neq 0$ and $\phi(x)$ is continuous at $x=2$, then in the neighbourhood of $x=2$
- (a) f is increasing if $\phi(2) < 0$ (b) f is decreasing if $\phi(2) > 0$
 (c) f is neither increasing nor decreasing (d) f is increasing if $\phi(2) > 0$
67. If $f(x) = x^3 - 6x^2 + ax + b$ is defined on $[1, 3]$ satisfies Rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$ then
- (a) $a = -11, b = 6$ (b) $a = -11, b = -6$ (c) $a = 11, b \in R$ (d) $a = 22, b = -6$
68. For which of the following function(s) Lagrange's mean value theorem is not applicable in $[1, 2]$?
- (a) $f(x) = \begin{cases} \frac{3}{2} - x, & x < \frac{3}{2} \\ \left(\frac{3}{2} - x\right)^2, & x \geq \frac{3}{2} \end{cases}$ (b) $f(x) = \begin{cases} \frac{\sin(x-1)}{x-1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$
 (c) $f(x) = (x-1)|x-1|$ (d) $f(x) = |x-1|$
69. If the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^2 = 16x$ intersect at right angles, then :
- (a) $a = \pm 1$ (b) $a = \pm \sqrt{3}$ (c) $a = \pm \frac{1}{\sqrt{3}}$ (d) $a = \pm \sqrt{2}$
70. If the line $x \cos \alpha + y \sin \alpha = P$ touches the curve $4x^3 = 27ay^2$, then $\frac{P}{a} =$
- (a) $\cot^2 \alpha \cos \alpha$ (b) $\cot^2 \alpha \sin \alpha$ (c) $\tan^2 \alpha \cos \alpha$ (d) $\tan^2 \alpha \sin \alpha$

Answers

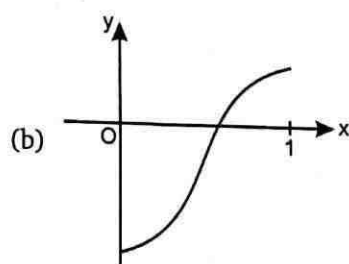
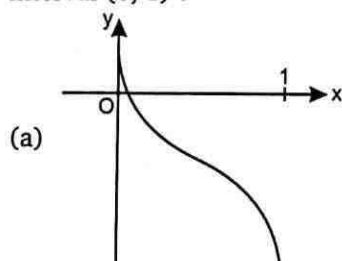
1. (d)	2. (b)	3. (a)	4. (c)	5. (b)	6. (c)	7. (d)	8. (d)	9. (c)	10. (b)
11. (c)	12. (c)	13. (c)	14. (c)	15. (c)	16. (a)	17. (b)	18. (a)	19. (b)	20. (b)
21. (b)	22. (d)	23. (d)	24. (a)	25. (d)	26. (c)	27. (b)	28. (d)	29. (c)	30. (d)
31. (b)	32. (c)	33. (d)	34. (d)	35. (a)	36. (b)	37. (c)	38. (a)	39. (c)	40. (a)
41. (a)	42. (b)	43. (b)	44. (a)	45. (d)	46. (b)	47. (a)	48. (a)	49. (d)	50. (a)
51. (b)	52. (d)	53. (b)	54. (a)	55. (a)	56. (c)	57. (b)	58. (c)	59. (b)	60. (c)
61. (b)	62. (a)	63. (b)	64. (c)	65. (b)	66. (d)	67. (c)	68. (a)	69. (d)	70. (a)

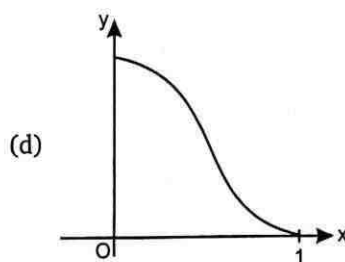
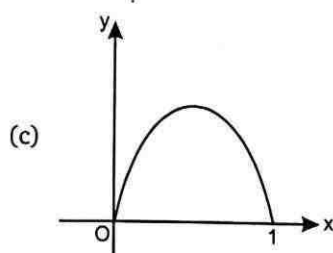
Exercise-2 : One or More than One Answer is/are Correct

- Common tangent(s) to $y = x^3$ and $x = y^3$ is/are :
 - $x - y = \frac{1}{\sqrt{3}}$
 - $x - y = -\frac{1}{\sqrt{3}}$
 - $x - y = \frac{2}{3\sqrt{3}}$
 - $x - y = \frac{-2}{3\sqrt{3}}$
- Let $f: [0, 8] \rightarrow \mathbb{R}$ be differentiable function such that $f(0) = 0$, $f(4) = 1$, $f(8) = 1$, then which of the following hold(s) good ?
 - There exist some $c_1 \in (0, 8)$ where $f'(c_1) = \frac{1}{4}$
 - There exist some $c \in (0, 8)$ where $f'(c) = \frac{1}{12}$
 - There exist $c_1, c_2 \in [0, 8]$ where $8f'(c_1)f(c_2) = 1$
 - There exist some $\alpha, \beta \in (0, 2)$ such that $\int_0^8 f(t) dt = 3(\alpha^2 f(\alpha^3) + \beta^2 f(\beta^3))$
- If $f(x) = \begin{cases} \sin^{-1}(\sin x) & x > 0 \\ \frac{\pi}{2} & x = 0 \\ \cos^{-1}(\cos x) & x < 0 \end{cases}$, then
 - $x = 0$ is a point of maxima
 - $f(x)$ is continuous $\forall x \in \mathbb{R}$
 - global maximum value of $f(x) \forall x \in \mathbb{R}$ is π
 - global minimum value of $f(x) \forall x \in \mathbb{R}$ is 0
- A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \begin{cases} x^4 \left(2 + \sin \frac{1}{x} \right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then
 - f has a continuous derivative $\forall x \in \mathbb{R}$
 - f is a bounded function
 - f has an global minimum at $x = 0$
 - f'' is continuous $\forall x \in \mathbb{R}$
- If $|f''(x)| \leq 1 \forall x \in \mathbb{R}$, and $f(0) = 0 = f'(0)$, then which of the following can not be true ?
 - $f\left(-\frac{1}{2}\right) = \frac{1}{6}$
 - $f(2) = -4$
 - $f(-2) = 3$
 - $f\left(\frac{1}{2}\right) = \frac{1}{5}$
- Let $f: [-3, 4] \rightarrow \mathbb{R}$ such that $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true ?
 - $f(x)$ has a relative minimum on $(-3, 4)$
 - $f(x)$ has a minimum on $[-3, 4]$
 - $f(x)$ has a maximum on $[-3, 4]$
 - if $f(3) = f(4)$, then $f(x)$ has a critical point on $[-3, 4]$

7. Let $f(x)$ be twice differentiable function such that $f''(x) > 0$ in $[0, 2]$. Then :
- $f(0) + f(2) = 2f(1)$, for atleast one $c, c \in (0, 2)$
 - $f(0) + f(2) < 2f(1)$
 - $f(0) + f(2) > 2f(1)$
 - $2f(0) + f(2) > 3f\left(\frac{2}{3}\right)$
8. Let $g(x)$ be a cubic polynomial having local maximum at $x = -1$ and $g'(x)$ has a local minimum at $x = 1$. If $g(-1) = 10, g(3) = -22$, then :
- perpendicular distance between its two horizontal tangents is 12
 - perpendicular distance between its two horizontal tangents is 32
 - $g(x) = 0$ has atleast one real root lying in interval $(-1, 0)$
 - $g(x) = 0$, has 3 distinct real roots
9. The function $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 2\lambda x + 5$ has a maximum and a minimum for :
- $\lambda \in (-4, \infty)$
 - $\lambda \in (-\infty, 0)$
 - $\lambda \in (-3, 3)$
 - $\lambda \in (1, \infty)$
10. The function $f(x) = 1 + x \ln\left(x + \sqrt{1+x^2}\right) - \sqrt{1-x^2}$ is :
- strictly increasing $\forall x \in (0, 1)$
 - strictly decreasing $\forall x \in (-1, 0)$
 - strictly decreasing for $x \in (-1, 0)$
 - strictly decreasing for $x \in (0, 1)$
11. Let m and n be positive integers and $x, y > 0$ and $x + y = k$, where k is constant. Let $f(x, y) = x^m y^n$, then :
- $f(x, y)$ is maximum when $x = \frac{mk}{m+n}$
 - $f(x, y)$ is maximum where $x = y$
 - maximum value of $f(x, y)$ is $\frac{m^n n^m k^{m+n}}{(m+n)^{m+n}}$
 - maximum value of $f(x, y)$ is $\frac{k^{m+n} m^m n^n}{(m+n)^{m+n}}$
12. The straight line which is both tangent and normal to the curve $x = 3t^2, y = 2t^3$ is :
- $y + \sqrt{3}(x-1) = 0$
 - $y - \sqrt{3}(x-1) = 0$
 - $y + \sqrt{2}(x-2) = 0$
 - $y - \sqrt{2}(x-2) = 0$
13. A curve is such that the ratio of the subnormal at any point to the sum of its co-ordinates is equal to the ratio of the ordinate of this point to its abscissa. If the curve passes through $(1, 0)$, then possible equation of the curve(s) is :
- $y = x \ln x$
 - $y = \frac{\ln x}{x}$
 - $y = \frac{2(x-1)}{x^2}$
 - $y = \frac{1-x^2}{2x}$

14. A parabola of the form $y = ax^2 + bx + c$ ($a > 0$) intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. The number of possible distinct intersection(s) of these graph can be :
 (a) 0 (b) 2 (c) 3 (d) 4
15. Gradient of the line passing through the point (2, 8) and touching the curve $y = x^3$, can be :
 (a) 3 (b) 6 (c) 9 (d) 12
16. The equation $x + \cos x = a$ has exactly one positive root, then :
 (a) $a \in (0, 1)$ (b) $a \in (2, 3)$ (c) $a \in (1, \infty)$ (d) $a \in (-\infty, 1)$
17. Given that $f(x)$ is a non-constant linear function. Then the curves :
 (a) $y = f(x)$ and $y = f^{-1}(x)$ are orthogonal
 (b) $y = f(x)$ and $y = f^{-1}(-x)$ are orthogonal
 (c) $y = f(-x)$ and $y = f^{-1}(x)$ are orthogonal
 (d) $y = f(-x)$ and $y = f^{-1}(-x)$ are orthogonal
18. Let $f(x) = \int_0^x e^{t^3} (t^2 - 1)t^2(t + 1)^{2011}(t - 2)^{2012}$ at ($x > 0$) then :
 (a) The number of point of inflections is atleast 1
 (b) The number of point of inflections is 0
 (c) The number of point of local maxima is 1
 (d) The number of point of local minima is 1
19. Let $f(x) = \sin x + ax + b$. Then $f(x) = 0$ has :
 (a) only one real root which is positive if $a > 1, b < 0$
 (b) only one real root which is negative if $a > 1, b > 0$
 (c) only one real root which is negative if $a < -1, b < 0$
 (d) only one real root which is positive if $a < -1, b < 0$
20. Which of the following graphs represent function whose derivatives have a maximum in the interval (0, 1) ?





21. Consider $f(x) = \sin^5 x + \cos^5 x - 1$, $x \in \left[0, \frac{\pi}{2}\right]$, which of the following is/are correct ?

- (a) f is strictly decreasing in $\left[0, \frac{\pi}{4}\right]$
- (b) f is strictly increasing in $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$
- (c) There exist a number 'c' in $\left(0, \frac{\pi}{2}\right)$ such that $f'(c) = 0$
- (d) The equation $f(x) = 0$ has only two roots in $\left[0, \frac{\pi}{2}\right]$

22. Let $f(x) = \begin{cases} x^{2\alpha+1} \ln x & ; x > 0 \\ 0 & ; x = 0 \end{cases}$

If $f(x)$ satisfies Rolle's theorem in interval $[0, 1]$, then α can be :

- (a) $-\frac{1}{2}$
- (b) $-\frac{1}{3}$
- (c) $-\frac{1}{4}$
- (d) -1

23. Which of the following is/are true for the function $f(x) = \int_0^x \frac{\cos t}{t} dt$ ($x > 0$) ?

- (a) $f(x)$ is monotonically increasing in $\left((4n-1)\frac{\pi}{2}, (4n+1)\frac{\pi}{2}\right) \forall n \in \mathbb{N}$
- (b) $f(x)$ has a local minima at $x = (4n-1)\frac{\pi}{2} \forall n \in \mathbb{N}$
- (c) The points of inflection of the curve $y = f(x)$ lie on the curve $x \tan x + 1 = 0$
- (d) Number of critical points of $y = f(x)$ in $(0, 10\pi)$ are 19

24. Let $F(x) = (f(x))^2 + (f'(x))^2$, $F(0) = 6$, where $f(x)$ is a thrice differentiable function such that $|f(x)| \leq 1 \forall x \in [-1, 1]$, then choose the correct statement(s)

- (a) there is atleast one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $|f'(x)| \leq 2$
- (b) there is atleast one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $F(x) \leq 5$
- (c) there is no point of local maxima of $F(x)$ in $(-1, 1)$
- (d) for some $c \in (-1, 1)$, $F(c) \geq 6$, $F'(c) = 0$ and $F''(c) \leq 0$

25. Let $f(x) = \begin{cases} x^3 + x^2 - 10x; & -1 \leq x < 0 \\ \sin x; & 0 \leq x < \frac{\pi}{2} \\ 1 + \cos x; & \frac{\pi}{2} \leq x \leq \pi \end{cases}$

then $f(x)$ has :

- (a) local maximum at $x = \frac{\pi}{2}$ (b) local minimum at $x = \frac{\pi}{2}$
 (c) absolute maximum at $x = 0$ (d) absolute maximum at $x = -1$

26. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to :

- (a) $\frac{\sqrt{2}}{4}$ (b) $\frac{3\sqrt{2}}{4}$ (c) $\frac{5\sqrt{2}}{4}$ (d) $\frac{7\sqrt{2}}{4}$

27. For the equation $\frac{e^{-x}}{1+x} = \lambda$ which of the following statement(s) is/are correct ?

- (a) When $\lambda \in (0, \infty)$ equation has 2 real and distinct roots
 (b) When $\lambda \in (-\infty, -e^2)$ equation has 2 real and distinct roots
 (c) When $\lambda \in (0, \infty)$ equation has 1 real root
 (d) When $\lambda \in (-e, 0)$ equation has no real root

28. If $y = mx + 5$ is a tangent to the curve $x^3y^3 = ax^3 + by^3$ at $P(1, 2)$, then

- (a) $a + b = \frac{18}{5}$ (b) $a > b$ (c) $a < b$ (d) $a + b = \frac{19}{5}$

29. If $(f(x) - 1)(x^2 + x + 1)^2 - (f(x) + 1)(x^4 + x^2 + 1) = 0$

$\forall x \in \mathbb{R} - \{0\}$ and $f(x) \neq \pm 1$, then which of the following statement(s) is/are correct ?

- (a) $|f(x)| \geq 2 \forall x \in \mathbb{R} - \{0\}$ (b) $f(x)$ has a local maximum at $x = -1$
 (c) $f(x)$ has a local minimum at $x = 1$ (d) $\int_{-\pi}^{\pi} (\cos x) f(x) dx = 0$

Answers

1.	(c, d)	2.	(a, c, d)	3.	(a, c)	4.	(a, c)	5.	(a, b, c, d)	6.	(b, c, d)
7.	(c, d)	8.	(b, d)	9.	(a, b, c, d)	10.	(a, c)	11.	(a, d)	12.	(c, d)
13.	(a, d)	14.	(b, c, d)	15.	(a, d)	16.	(b, c)	17.	(b, c)	18.	(a, d)
19.	(a, b, c)	20.	(a, b)	21.	(a, b, c, d)	22.	(b, c)	23.	(a, b, c)	24.	(a, b, d)
25.	(a, d)	26.	(b)	27.	(b, c, d)	28.	(a, d)	29.	(a, b, c, d)		

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $y = f(x)$ such that $xy = x + y + 1$, $x \in \mathbb{R} - \{-1\}$ and $g(x) = xf(x)$

- The minimum value of $g(x)$ is :
 (a) $3 - \sqrt{2}$ (b) $3 + \sqrt{2}$ (c) $3 - 2\sqrt{2}$ (d) $3 + 2\sqrt{2}$
- There exists two values of x, x_1 and x_2 where $g'(x) = \frac{1}{2}$, then $|x_1| + |x_2| =$
 (a) 1 (b) 2 (c) 4 (d) 5

Paragraph for Question Nos. 3 to 5

Let $f(x) = \begin{cases} 1-x & ; 0 \leq x \leq 1 \\ 0 & ; 1 < x \leq 2 \\ (2-x)^2 & ; 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t) dt$.

Let the tangent to the curve $y = g(x)$ at point P whose abscissa is $\frac{5}{2}$ cuts x -axis in point Q .

Let the perpendicular from point Q on x -axis meets the curve $y = g(x)$ in point R .

- $g(1) =$
 (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2
- Equation of tangent to the curve $y = g(x)$ at P is :
 (a) $3y = 12x + 1$ (b) $3y = 12x - 1$ (c) $12y = 3x - 1$ (d) $12y = 3x + 1$
- If θ be the angle between tangents to the curve $y = g(x)$ at point P and R ; then $\tan \theta$ equals to :
 (a) $\frac{5}{6}$ (b) $\frac{5}{14}$ (c) $\frac{5}{7}$ (d) $\frac{5}{12}$

Paragraph for Question Nos. 6 to 8

Let $f(x) < 0 \forall x \in (-\infty, 0)$ and $f(x) > 0 \forall x \in (0, \infty)$ also $f(0) = 0$. Again $f'(x) < 0 \forall x \in (-\infty, -1)$ and $f'(x) > 0 \forall x \in (-1, \infty)$ also $f'(-1) = 0$ given $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = \infty$ and function is twice differentiable.

- If $f''(x) > 0 \forall x \in (-1, \infty)$ and $f'(0) = 1$ then number of solutions of equation $f(x) = x$ is :
 (a) 2 (b) 3 (c) 4 (d) None of these
- If $f''(x) < 0 \forall x \in (0, \infty)$ and $f'(0) = 1$ then number of solutions of equation $f(x) = x^2$ is :
 (a) 1 (b) 2 (c) 3 (d) 4

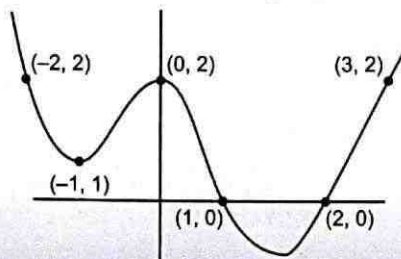
8. The minimum number of points where $f''(x)$ is zero is :

- (a) 1 (b) 2 (c) 3 (d) 4

Paragraph for Question Nos. 9 to 11

In the given figure graph of :

$$y = p(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n \text{ is given.}$$



9. The product of all imaginary roots of $p(x) = 0$ is :

- (a) -2 (b) -1 (c) $-1/2$ (d) none of these

10. If $p(x) + k = 0$ has 4 distinct real roots $\alpha, \beta, \gamma, \delta$ then $[\alpha] + [\beta] + [\gamma] + [\delta]$, (where $[\cdot]$ denotes greatest integer function) is equal to :

- (a) -1 (b) -2 (c) 0 (d) 1

11. The minimum number of real roots of equation $(p'(x))^2 + p(x)p''(x) = 0$ are :

- (a) 3 (b) 4 (c) 5 (d) 6

Paragraph for Question Nos. 12 to 14

The differentiable function $y = f(x)$ has a property that the chord joining any two points $A(x_1, f(x_1))$ and $B(x_2, f(x_2))$ always intersects y-axis at $(0, 2x_1x_2)$. Given that $f(1) = -1$, then :

12. $\int_0^{1/2} f(x) dx$ is equal to :

- (a) $\frac{1}{6}$ (b) $\frac{1}{8}$ (c) $\frac{1}{12}$ (d) $\frac{1}{24}$

13. The largest interval in which $f(x)$ is monotonically increasing, is :

- (a) $\left(-\infty, \frac{1}{2}\right]$ (b) $\left[\frac{-1}{2}, \infty\right)$ (c) $\left(-\infty, \frac{1}{4}\right]$ (d) $\left[\frac{-1}{4}, \infty\right)$

14. In which of the following intervals, the Rolle's theorem is applicable to the function $F(x) = f(x) + x$?

- (a) $[-1, 0]$ (b) $[0, 1]$ (c) $[-1, 1]$ (d) $[0, 2]$

Paragraph for Question Nos. 15 to 16

Let $f(x) = 1 + \int_0^1 (xe^y + ye^x) f(y) dy$ where x and y are independent variables.

15. If complete solution set of 'x' for which function $h(x) = f(x) + 3x$ is strictly increasing is $(-\infty, k)$ then $\left[\frac{4}{3}e^k\right]$ equals to : (where $[\cdot]$ denotes greatest integer function):

(a) 1 (b) 2 (c) 3 (d) 4

16. If acute angle of intersection of the curves $\frac{x}{2} + \frac{y}{3} + \frac{1}{3} = 0$ and $y = f(x)$ be θ then $\tan \theta$ equals to :

(a) $\frac{8}{25}$ (b) $\frac{16}{25}$ (c) $\frac{14}{25}$ (d) $\frac{4}{5}$

Answers

1. (d)	2. (c)	3. (b)	4. (c)	5. (b)	6. (d)	7. (b)	8. (a)	9. (d)	10. (a)
11. (b)	12. (d)	13. (c)	14. (b)	15. (c)	16. (a)				

**Exercise-4 : Matching Type Problems**

1. Column-I gives pair of curves and column-II gives the angle θ between the curves at their intersection point.

Column-I		Column-II	
(A)	$y = \sin x, y = \cos x$	(P)	$\frac{\pi}{4}$
(B)	$x^2 = 4y, y = \frac{8}{x^2 + 4}$	(Q)	$\frac{\pi}{2}$
(C)	$\frac{x^2}{18} + \frac{y^2}{8} = 1, x^2 - y^2 = 5$	(R)	$\tan^{-1} 3$
(D)	$xy = 1, x^2 - y^2 = 5$	(S)	$\tan^{-1} 5$
		(T)	$\tan^{-1}(2\sqrt{2})$

2.

Column-I		Column-II	
(A)	$(\sin^{-1} x)^{\cos^{-1} x} - (\cos^{-1} x)^{\sin^{-1} x} \forall x \in (\cos 1, \sin 1)$	(P)	Always positive
(B)	$(\cos x)^{\sin x} - (\sin x)^{\cos x} \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	(Q)	Always negative
(C)	$(\sin x)^{\sin x} - (\cos x)^{\sin x} \forall x \in \left(0, \frac{\pi}{2}\right)$	(R)	May be positive or negative for some values of x
(D)	$(\ln(\ln x))^{\ln(\ln x)} - (\ln x)^{\ln x} \forall x \in (e^e, \infty)$	(S)	May result in zero for some of values of x
		(T)	Indeterminate

3. Let $f(x) = \frac{x^3 - 4}{(x-1)^3} \forall x \neq 1$, $g(x) = \frac{x^4 - 2x^2}{4} \forall x \in \mathbb{R}$, $h(x) = \frac{x^3 + 4}{(x+1)^3} \forall x \neq -1$,

Column-I		Column-II	
(A)	The number of possible distinct real roots of equation $f(x) = c$ where $c \geq 4$ can be	(P)	0
(B)	The number of possible distinct real roots of equation $g(x) = c$, where $c \geq 0$ can be	(Q)	1
(C)	The number of possible distinct real roots of equation $h(x) = c$, where $c \geq 1$ can be	(R)	2

(D)	The number of possible distinct real roots of equation $g(x) = c$ where $-1 < c < 0$ can be	(S)	3
		(T)	4

4.

	Column-I		Column-II
(A)	If α, β, γ are roots of $x^3 - 3x^2 + 2x + 4 = 0$ and $y = 1 + \frac{\alpha}{x-\alpha} + \frac{\beta x}{(x-\alpha)(x-\beta)} + \frac{\gamma x^2}{(x-\alpha)(x-\beta)(x-\gamma)}$ then value of y at $x = 2$ is :	(P)	2
(B)	If $x^3 + ax + 1 = 0$ and $x^4 + ax + 1 = 0$ have a common roots then the value of $ a $ can be equal to	(Q)	3
(C)	The number of local maximas of the function $x^2 + 4\cos x + 5$ is more than	(R)	4
(D)	If $f(x) = 2 x ^3 + 3x^2 - 12 x + 1$, where $x \in [-1, 2]$ then greatest value of $f(x)$ is more than	(S)	5
		(T)	0

5.

	Column-I		Column-II
(A)	Maximum value of $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$	(P)	0
(B)	The value of $\left[4 \sum_{n=1}^{\infty} \cot^{-1} \left(1 + \sum_{k=1}^n 2k \right) \right] =$ ([.] represent greatest integer function)	(Q)	1
(C)	Let $f(x) = x \sin \pi x$, $x > 0$ then number of points in $(0, 2)$ where $f'(x)$ vanishes, is	(R)	2
(D)	$\lim_{x \rightarrow 0^+} \left[\frac{x}{e^x - 1} \right] =$ ([.] represent greatest integer function)	(S)	3

6. Consider the function $f(x) = \frac{\ln x}{8} - ax + x^2$ and $a \geq 0$ is a real constant :

Column-I		Column-II	
(A)	$f(x)$ gives a local maxima at	(P)	$a = 1; x = \frac{1}{4}$
(B)	$f(x)$ gives a local minima at	(Q)	$a > 1; x = \frac{a - \sqrt{a^2 - 1}}{4}$
(C)	$f(x)$ gives a point of inflection for	(R)	$0 \leq a < 1$
(D)	$f(x)$ is strictly increasing for all $x \in \mathbb{R}^+$	(S)	$a > 1; x = \frac{a + \sqrt{a^2 - 1}}{4}$

7. The function $f(x) = \sqrt{ax^3 + bx^2 + cx + d}$ has its non-zero local minimum and maximum values at $x = -2$ and $x = 2$ respectively. If 'a' is one of the root of $x^2 - x - 6 = 0$, then match the following :

Column-I		Column-II	
(A)	The value of 'a' is	(P)	0
(B)	The value of 'b' is	(Q)	24
(C)	The value of 'c' is	(R)	Greater than 32
(D)	The value of 'd' is	(S)	-2

8.

Column-I		Column-II	
(A)	The ratio of altitude to the radius of the cylinder of maximum volume that can be inscribed in a given sphere is	(P)	$\frac{1}{\sqrt{2}}$
(B)	The ratio of radius to the altitude of the cone of the greatest volume which can be inscribed in a given sphere is	(Q)	$\sqrt{2}$
(C)	The cone circumscribing the sphere of radius 'r' has the maximum volume if its semi vertical angle is θ , then $33 \sin \theta =$	(R)	$\frac{32}{3}$
(D)	The greatest value of $x^3 y^4$ if $2x + 3y = 7$, $x \geq 0, y \geq 0$ is	(S)	11

Answers

1. $A \rightarrow T$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow Q$
2. $A \rightarrow R, S$; $B \rightarrow Q$; $C \rightarrow R, S$; $D \rightarrow Q$
3. $A \rightarrow Q, R$; $B \rightarrow R, S$; $C \rightarrow Q, R, S$; $D \rightarrow P, R, T$
4. $A \rightarrow P$; $B \rightarrow P$; $C \rightarrow T$; $D \rightarrow P, Q, R, T$
5. $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow P$
6. $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow P$; $D \rightarrow R$
7. $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow R$
8. $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow R$

Exercise-5 : Subjective Type Problems

1. A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is A_1 and the area of the given sheet is A_2 . If $\frac{A_2}{A_1} = m + \sqrt{n}$, where $m, n \in \mathbb{N}$, then $m + n$ is equal to.
2. On $[1, e]$, the least and greatest values of $f(x) = x^2 \ln x$ are m and M respectively, then $[\sqrt{M + m}]$ is : (where $[\]$ denotes greatest integer function)
3. If $f(x) = \frac{px}{e^x} - \frac{x^2}{2} + x$ is a decreasing function for every $x \leq 0$. Find the least value of p^2 .
4. Let $f(x) = \begin{cases} xe^{ax} & , x \leq 0 \\ x + ax^2 - x^3 & , x > 0 \end{cases}$. Where a is a positive constant. The interval in which $f'(x)$ is increasing is $\left[\frac{k}{a}, \frac{a}{l}\right]$. Then $k + l$ is equal to
5. Find sum of all possible values of the real parameter ' b ' if the difference between the largest and smallest values of the function $f(x) = x^2 - 2bx + 1$ in the interval $[0, 1]$ is 4.
6. Let ' θ ' be the angle in radians between the curves $\frac{x^2}{36} + \frac{y^2}{4} = 1$ and $x^2 + y^2 = 12$. If $\theta = \tan^{-1}\left(\frac{a}{\sqrt{3}}\right)$; Find the value of a .
7. Let set of all possible values of λ such that $f(x) = e^{2x} - (\lambda + 1)e^x + 2x$ is monotonically increasing for $\forall x \in \mathbb{R}$ is $(-\infty, k]$. Find the value of k .
8. Let a, b, c and d be non-negative real number such that $a^5 + b^5 \leq 1$ and $c^5 + d^5 \leq 1$. Find the maximum value of $a^2c^3 + b^2d^3$.
9. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -8/x$, where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points respectively, then find the value of $(p + r)$.
10. $f(x) = \max |2 \sin y - x|$ where $y \in \mathbb{R}$ then determine the minimum value of $f(x)$.
11. Let $f(x) = \int_0^x ((a-1)(t^2 + t + 1)^2 - (a+1)(t^4 + t^2 + 1)) dt$. Then the total number of integral values of ' a ' for which $f'(x) = 0$ has no real roots is
12. The number of real roots of the equation $x^{2013} + e^{2014x} = 0$ is
13. Let the maximum value of expression $y = \frac{x^4 - x^2}{x^6 + 2x^3 - 1}$ for $x > 1$ is $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the value of $(p + q)$.

14. The least positive value of the parameter 'a' for which there exists atleast one line that is tangent to the graph of the curve $y = x^3 - ax$, at one point and normal to the graph at another point is $\frac{p}{q}$; where p and q are relatively prime positive integers. Find product pq.
15. Let $f(x) = x^2 + 2x - t^2$ and $f(x) = 0$ has two roots $\alpha(t)$ and $\beta(t)$ ($\alpha < \beta$) where t is a real parameter. Let $I(t) = \int_{\alpha}^{\beta} f(x) dx$. If the maximum value of $I(t)$ be λ and $|\lambda| = \frac{p}{q}$ where p and q are relatively prime positive integers. Find the product (pq).
16. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of salt runs into the tank at the rate of 1 lit/min. The homogenised mixture is pumped out of the tank at the rate of 3 lit/min. If T be the time when the amount of salt in the tank is maximum. Find [T] (where [.] denotes greatest integer function)
17. If $f(x)$ is continuous and differentiable in $[-3, 9]$ and $f'(x) \in [-2, 8] \forall x \in (-3, 9)$. Let N be the number of divisors of the greatest possible value of $f(9) - f(-3)$, then find the sum of digits of N.
18. It is given that $f(x)$ is defined on R satisfying $f(1) = 1$ and for $\forall x \in R$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 1 - x$, then $g(2002) =$
19. The number of normals to the curve $3y^3 = 4x$ which passes through the point (0, 1) is
20. Find the number of real root(s) of the equation $ae^x = 1 + x + \frac{x^2}{2}$; where a is positive constant.
21. Let $f(x) = ax + \cos 2x + \sin x + \cos x$ is defined for $\forall x \in R$ and $a \in R$ and is strictly increasing function. If the range of a is $\left[\frac{m}{n}, \infty\right)$, then find the minimum value of $(m - n)$.
22. If p_1 and p_2 are the lengths of the perpendiculars from origin on the tangent and normal drawn to the curve $x^{2/3} + y^{2/3} = 6^{2/3}$ respectively. Find the value of $\sqrt{4p_1^2 + p_2^2}$.

Answers

1.	9	2.	2	3.	1	4.	1	5.	1	6.	2	7.	3
8.	1	9.	5	10.	2	11.	3	12.	1	13.	7	14.	12
15.	12	16.	27	17.	3	18.	1	19.	1	20.	1	21.	9
22.	6												

